

Executive Function Measures for Children: A Scoping Review of Ecological Validity

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Abstract

Children use executive function (EF) skills within everyday occupations; however, EF poses a difficult and complex construct to measure. Currently, many measures of EF lack applicability to daily life, or ecological validity. The aim of this scoping review was to examine two aspects of ecological validity across measures, assessments, and tasks of EF in children. A scoping review of 355 peer-reviewed articles published between 1996 and 2016 was performed. Searching revealed 43 articles addressing the ecological validity of EF measures for children and 40 measures addressing ecological validity. An increasing number of articles address ecological validity of EF measures. Future research should address the interplay between context and EF performance. In addition, research should begin recognizing the importance of parental involvement in assessments, as well as ways to capture the EF strengths of children.

Keywords

children, executive function, scoping reviews, ecological validity, measurement

Introduction

Children use executive functions to complete everyday activities; however, higher order processes pose a difficult construct to measure, partially due to the wide range of skills encompassed by executive function (EF; e.g., inhibition, working memory, shifting, and planning; Baron, 2007; Chaytor & Schmitter-Edgecombe, 2003; Kenworthy, Yerys, Anthony, & Wallace, 2008). Furthermore, researchers argue that current methods of measuring EF lack applicability to daily life or ecological validity (Burgess, Alderman, Evans, Emslie, & Wilson, 1998; Kenworthy et al., 2008). Therefore, this scoping review examines evidence related to the ecological validity of current tasks and behavioral assessments used to measure aspects of EF in children. In addition, we conclude by highlighting the role of occupational therapy (OT) principles and perspectives in the context of the ecological validity of EF measures for children.

Ecological validity refers to the extent to which an assessment produces logically sound data representing individuals' interactions with their surroundings. Specifically, Chaytor and Schmitter-Edgecombe (2003) defined the concept as the degree to which results obtained through experiments and assessments are related to those obtained in authentic contexts. Furthermore, components of "representativeness" (the extent to which an assessment corresponds to situations outside the lab or clinic) and "generalizability" (the degree to which concerns on the assessment are concerns in everyday life) were added to the definition (Burgess et al., 2006

adapted from Kvavilashvili & Ellis, 2004). Ecological validity is related to how an assessment provides clinical utility beyond diagnostic utility.

There are two approaches to identify the degree of ecological validity of a measure: verisimilitude and veridicality (Chaytor & Schmitter-Edgecombe, 2003; Kenworthy et al., 2008). Verisimilitude reflects the extent to which the theoretical foundations of an assessment mirror cognitive demands in everyday life, which typically requires the creation of new assessments with ecological goals in mind from the beginning. Verisimilitude is not a measure of discriminant validity (i.e., discriminate between those with and without a diagnosis); rather, it represents how EF aligns with tasks in real life. Therefore, assessments with a high degree of verisimilitude typically are more sensitive to performance change as one increases functional skills.

Veridicality relates to the extent to which existing assessments are empirically related to measures of everyday life. Typically, statistical analyses are used to relate the performance of a traditional assessment to a real-world

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Table 1. Search Terms.

Population	Behavior	Property	Type
Children, pediatric, kid	Executive function, executive control, cognitive control, inhibition, planning, shifting, cognitive flexibility, working memory	Ecological validity, real life, real world, authentic context, natural context, authentic environment, natural environment, everyday life	Assessment, task, paradigm, measure

assessment (Chaytor & Schmitter-Edgecombe, 2003). Although some tests were not designed to test cognition as it pertains to everyday function, these assessments may still have value and predictive abilities of everyday function. In psychometric terms, veridicality is related to concurrent validity (i.e., two measures reflecting the same incidences of behavior; Portney & Watkins, 2009), but refers solely to a degree of ecological validity. Thus, verisimilitude and veridicality provide distinct methods to further understand the complexities of ecological validity across measures of EF.

This scoping review examined two aspects of ecological validity (i.e., veridicality and verisimilitude) across measures, assessments, and tasks of EF in children (2-12 years old). We addressed the following research question:

Research Question 1: What do we know from the literature about the ecological validity of current tasks and behavioral measures examining EF in children?

Method

We conducted a scoping review to understand the breadth and depth of ecologically valid measures for children ages 2 to 12 years. A scoping review aims to examine the extent and range of research regarding our topic, allowing us to map key concepts and understand research gaps surrounding ecological validity (Arksey & O'Malley, 2005). We followed a five-step process of conducting a scoping review (Arksey & O'Malley, 2005; Levac, Colquhoun, & O'Brien, 2010), which included (a) identifying the research question, (b) identifying relevant studies, (c) study selection, (d) charting the data, and (e) collating, summarizing, and reporting results.

Identifying Relevant Studies

We searched PubMed and PsycInfo from 1996 to September 2016. See Table 1 for search terms. To ensure that we were thorough in the search process, we conducted a forward and backward hand search by examining the reference lists as well as articles citing an included article. We also searched the reference lists of other systematic reviews to provide a comprehensive scoping review (Arksey & O'Malley, 2005).

Inclusion. We included studies providing evidence of the ecological validity or everyday life application of assessments, tasks, or measures of EF. In addition, we included studies that

measured global EF or one aspect of EF (e.g., only inhibition). Finally, we included studies of children aged 2 to 12 years to understand measures associated with early and middle childhood. Although executive function across childhood varies due to development, many assessments also span a wide range of childhood (Anderson, 2002; Kenworthy et al., 2008).

Exclusion. We excluded intervention studies, pharmacological studies, dissertations, theses, book chapters, and reviews. We also excluded IQ assessments or studies focused solely on language processing as well as studies that used neuroimaging methods to test EF or focused on neural mechanisms of EF. Finally, studies pertaining to nonhuman subjects and studies not written in English were excluded.

Study Selection

PubMed ($n = 281$) and PsycInfo ($n = 166$) yielded a total of 447 articles, and when duplicates were removed ($n = 92$ duplicates), the search yielded a total of 355 articles. A review of 100% of abstracts occurred between two of the authors to assure reliability in selection. Reviewers tested percentage agreement with 73.5% of the articles and achieved 95.79% agreement; the reviewers discussed the remaining 4.21% of articles and agreed on inclusion or exclusion.

Results

Characteristics of Research Studies

After the selection process, 43 articles matched review criteria. Approximately 86.05% ($n = 37$ studies) occurred within the last 10 years, showing a recent surge of literature aimed toward creating and understanding the ecological validity of measures of EF. Articles on the topic of interest were found in 23 journals. In addition, 42 different EF measures for children aged 2 to 12 years were examined within the studies; six measures were parent, caregiver, or teacher questionnaires, an additional questionnaire occurred with a battery of EF tasks, and 35 measures were clinician-rated behavioral tasks. See Figure 1 for data chart.

Populations

Studies examined a wide range of conditions, including attention deficit hyperactive disorder (ADHD, $n = 12$), typical

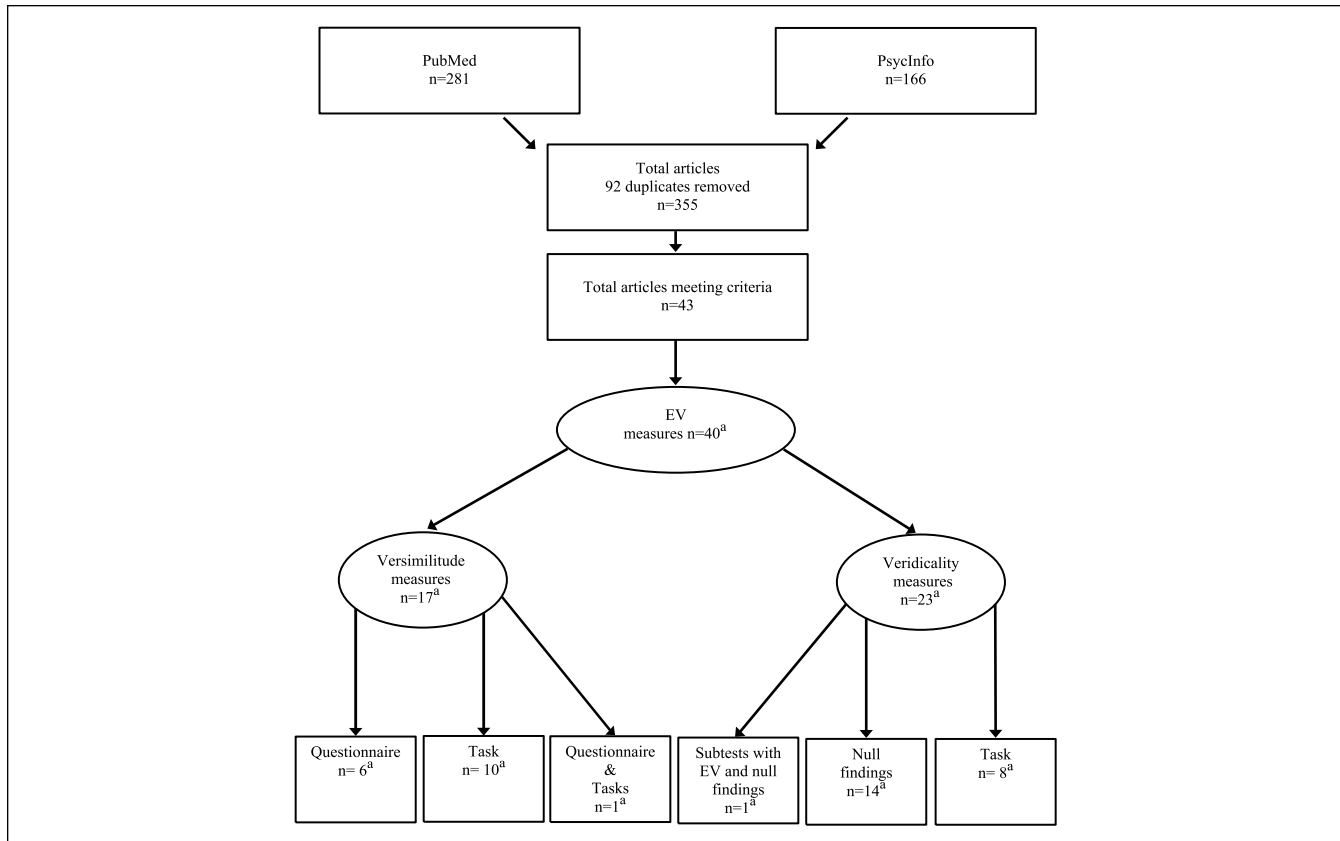


Figure 1. Step 4: Charting the data.

Note. EV = ecological validity.

^aRefers to measures instead of articles.

development ($n = 11$), autism spectrum disorder (ASD, $n = 7$), traumatic brain injury (TBI, $n = 5$), brain tumors ($n = 4$), epilepsy/seizures ($n = 2$), schizophrenia ($n = 1$), developmental coordination disorder (DCD, $n = 1$), congenital heart disease ($n = 1$), learning disorders ($n = 1$), spina bifida ($n = 1$), Down syndrome ($n = 1$), cerebral palsy (CP, $n = 1$), and children born very preterm ($n = 1$). Some studies included more than one population.

Versimilitude: Measures Theoretically Aligning With Everyday Function

Approximately 17 of the measures (42.5%) utilized a versimilitude approach (see Table 2). We determined whether a measure used a versimilitude approach if the measure's description outlined attributes of ecological validity, and if the measure was not adapted from another task. All parent, caregiver, or teacher report questionnaires used a versimilitude approach ($n = 7$ measures) and aimed to understand EF behavior in real life. Two of the measures (i.e., Children's Cooking Task, Do Eat Assessment) used daily living activities to assess EF. Another assessment (i.e., Rivermead Behavioral Memory Test for Children)

addressed memory relating to daily events (e.g., appointments, delivering a message). One measure utilized virtual reality to create a more real-world context to assess executive function (Jaroslawska, Gathercole, Logie, & Holmes, 2016). The remaining seven measures utilized different search, sorting, navigating, instruction following, and probability tasks.

Veridicality: Measures Relating to Everyday Life

Approximately 23 of the measures (57.5%) used a veridicality approach (see Table 3). We determined whether a measure used a veridicality approach if the measure was either a traditional measure adapted to exhibit more ecologically valid attributes, or if the traditional measure was compared with a versimilitude measure. In five studies, the Behavior Rating Inventory of Executive Function (BRIEF), which is considered an ecologically valid measure, was used to compare against traditional measures such as the Rey–Osterrieth Complex Figure Task (Davies, Field, Andersen, & Pestell, 2011); Biber Cognitive Estimations Test (MacAllister, Vasserman, Coulehan, Hall, & Bender, 2016); Clinical Evaluation of Language Fundamentals,

Table 2. Measures With Versimilitude Evidence.

Measure	Articles	Age in years	Diagnostic population	Format
Behavioral Assessment of Dysexecutive Syndrome–Child (BADS-C)	Baron (2007); de Almeida, Macedo, Lopes, and Monteiro (2014); Gilboa, Rosenblum, Fattal-Valevski, Toledano-Alhadeff, and Josman (2014); Roy, Allain, Roulin, Fournet, and Le Gall (2015); Siu and Zhou (2014); White, Burgess, and Hill (2009); Williams, Mazefsky, Walker, Minshew, and Goldstein (2014)	8-15	Developmental and neurological disorders, ADHD, ASD	Task and questionnaire
Rivermead Behavioral Memory Test for Children	Chen, Tsai, Hsu, Ma, and Lai (2013)	5-10	DCD	Task
The Children's Cooking Task	Chevignard, Catroppa, Galvin, and Anderson (2010)	8-20	TBI	Task
Iowa Gambling Task–Child	Crone and van der Molen (2004)	6-25	TD	Task
BRIEF	Fournet et al. (2015); Gerstle, Beebe, Drotar, Cassidy, and Marino (2016); Gilboa et al. (2014); Rosenthal et al. (2013); van den Bergh, Scheeren, Begeer, Koot, and Geurts (2014); Wehrle et al. (2016); Whittingham, Bodimeade, Lloyd, and Boyd (2014); Wolfe et al. (2013)	5-18	TBI, ASD, brain tumors, CHD, children born very preterm	Questionnaire
BRIEF–Preschool (BRIEF-P)	Isquith, Gioia, and Espy (2004); Pritchard, Kalback, McCurdy, and Capone (2015); Smithson et al. (2013)	2-5	TD, DS	Questionnaire
BRIEF–Self-Report (BRIEF-SR)	Li, Cao, Shao, and Xue (2014)	9-20	Schizophrenia	Questionnaire
Children's Executive Function Scale	Goulden and Silver (2009)	10-12	ADHD, learning disorders, epilepsy, TBI, HIV, brain tumors, TD, developmental disorders	Questionnaire
Real-world following-instructions task	Jaroslawska, Gathercole, Logie, and Holmes (2016)	7-11	TD	Task
Learning, Executive, and Attention Functioning Scale	Kronenberger, Castellanos, and Pisoni (2016)	6-17	Spina bifida	Questionnaire
Route Tasks	Lawrence et al. (2002, 2004)	6-12	ADHD	Task
Key Search Task	Lui and Tannock (2007)	7-12	TD	Task
Battersea Multitask Paradigm	Mackinlay, Charman, and Karmiloff-Smith (2006)	12	ASD	Task
Do Eat Assessment	Rosenblum, Frisch, Deutsch-Castel, and Josman (2015)	5-9	ADHD	Task
Ball-Search Field Task	Rosetti et al. (2016)	5-12	ADHD	Task
Cognition and Motivation in Everyday Life	Van Liefveringe et al. (2016)	6-16	ADHD	Questionnaire
Modified Hayling Sentence Completion Task	White et al. (2009)	8-46	ASD	Task

Note. ADHD = attention deficit hyperactive disorder; ASD = autism spectrum disorder; DCD = developmental coordination disorder; TBI = traumatic brain injury; TD = typically developing; CHD = congenital heart disease; BRIEF = Behavior Rating Inventory of Executive Function; DS = Down syndrome.

NEPSY Visual Attention, and Willoughby Computerized Battery of Executive Functioning Task (Tamm, Brenner, Bamberger, & Becker, 2016); Children's Category Test; Controlled Oral Word Association Test; Twoer; Test of Everyday Attention for Children (Payne, Hyman, Shores, & North, 2011); Trail Making Test; and Wisconsin Card Sort Task (Vriezen & Pigott, 2002). Other studies used diagnostic rating scales of behavior (i.e., ADHD or ASD), observations, and teacher ratings to test alignment with traditional EF tasks (Floyd & Kirby, 2001; Pnevmatikos & Trikkaliotis, 2013; Solanto et al., 2001; Weis & Totten, 2004). Some studies adapted aspects of traditional measures to increase ecological validity. Specifically, de Vries and Geurts (2012) modified a traditional switch task to create a more complex task representing everyday life, and Diaz-Orueta et al. (2014) and Nolin, Martin, & Bouchard (2009) utilized virtual reality to replace a laboratory context with a natural school context to perform an EF task.

Discussion

The purpose of this scoping review was to examine the literature related to the ecological validity of current tasks and behavioral assessments used to measure aspects of EF in children. The increase in studies within the last 10 years demonstrates an emerging importance on testing the ecological validity of measures of EF for children. Findings suggest that the ecological validity of measures were tested on a wide variety of populations for EF differences, specifically 14 populations. Of these populations, children with ADHD, ASD, or a TBI were the most widely examined diagnostic groups, clearly illustrating the significance of understanding EF in the daily lives of children.

Our findings showed the BRIEF/BRIEF-P/BRIEF-SR ($n = 17$ studies) and the BADS-C ($n = 7$ studies) as the most widely cited assessments throughout this review, which aligns with a previous review by Kenworthy et al.

Table 3. Measures With Evidence of Veridicality.

Measure	Articles	Age in years	Diagnostic population	Format	Ecological validity examination
Rey–Osterrieth Complex Figure	Davies, Field, Andersen, and Pestell (2011)	6-17	Brain injuries	Task	BRIEF ^a
Gender Emotion Switch Task	de Vries and Geurts (2012)	8-12	ASD	Task	Adapted a traditional switch task
Virtual Continuous Performance Test	Díaz-Orueta et al. (2014); Nolin, Martin, and Bouchard (2009)	6-16	ADHD	Task	Adapted the Conner's Continuous Performance Task
Tower Go/NoGo Task	Floyd and Kirby (2001)	3-5	TD	Task	BASC ^a
Pinball Go/NoGo Task	Floyd and Kirby (2001)	3-5	TD	Task	BASC
Dog and Dragon Go/NoGo Task	Floyd and Kirby (2001)	3-5	TD	Task	BASC ^a
Snack Delay	Floyd and Kirby (2001)	3-5	TD	Task	BASC
Tongue Delay	Floyd and Kirby (2001)	3-5	TD	Task	BASC
Shapes Stroop	Floyd and Kirby (2001)	3-5	TD	Task	BASC
Children's Category Test	Payne, Hyman, Shores, and North (2011)	6-16	Brain tumor	Task	BRIEF ^a
Controlled Oral Word Association Test	Payne et al. (2011)	6-16	Brain tumor	Task	BRIEF ^a
Tower	Payne et al. (2011)	6-16	Brain tumor	Task	BRIEF ^a
Test of Everyday Attention for Children	Payne et al. (2011)	6-16	Brain tumor	Task	BRIEF ^b
Biber Cognitive Estimations Test	MacAllister, Vasserman, Coulehan, Hall, and Bender (2016)	7-16	Epilepsy	Task	BRIEF ^a
Go/NoGo Emotions	Pnevmatikos and Trikkaliotis (2013)	8-12	TD	Task	Adapted from a traditional go/no-go
Stop Signal Paradigm	Solanto et al. (2001)	7-10	ADHD	Task	Swanson, Nolan and Pelham- IV (SNAP-IV), ^a Classroom Observation Code, Conner's Parent and Teacher Questionnaires ^a
Choice-Delay Task	Solanto et al. (2001)	7-10	ADHD	Task	SNAP-IV, Classroom Observation Code, Conner's Parent and Teacher Questionnaires
Clinical Evaluation of Language Fundamentals	Tamm, Brenner, Bamberger, and Becker (2016)	4	ADHD	Task	BRIEF-P ^a
A Developmental NEuroPSYchological Assessment (NEPSY), Visual Attention	Tamm et al. (2016)	4	ADHD	Task	BRIEF-P ^a
Willoughby Computerized Battery of Executive Functioning Tasks	Tamm et al. (2016)	4	ADHD	Task	BRIEF-P ^a
Trail Making Test	Vriezen and Pigott (2002)	4-14	TBI	Task	BRIEF ^a
Wisconsin Cart Sort Task	Vriezen and Pigott (2002)	4-14	TBI	Task	BRIEF ^a
Continuous Performance Test-II	Weis and Totten (2004)	6-8	TD, ADHD	Task	Analogue task coded observation, ^a classroom observations, ^a Conners' Parent and Teacher Rating scales ^a

Note. BRIEF = Behavior Rating Inventory of Executive Function; ASD = autism spectrum disorder; ADHD = attention deficit hyperactive disorder; TD = typically developing; BASC = Behavioral Assessment System for Children; TBI = traumatic brain injury.

^aNull findings.

^bPartial null findings.

(2008). The BRIEF and the BADS-C both utilize questionnaires, and provide standardized developmental norms of executive function in children (Kenworthy et al., 2008). Thus, when reviewed measures assessed the veridicality of a traditional assessment, many utilized correlations with

the BRIEF. However, these veridicality measures were all found to have null associations with the BRIEF ($n = 10$), except one measure demonstrated partial null findings (i.e., some subtests showed veridicality and others were null). Other authors discuss null findings with the BRIEF

(e.g., McAuley, Chen, Goos, Schachar, & Crosbie, 2010) and propose it is unclear if null measures truly lack ecological validity, or if differences in measurement format reduces the potential for ecological validity (i.e., parent ratings vs. a task; Chevignard, Soo, Galvin, Catroppa, & Eren, 2012; McAuley et al., 2010).

Although parent ratings provide pertinent information and insight into everyday child behavior, many issues arise when relying heavily on parent ratings to determine the ecological validity of tasks. Research indicates inconsistencies on rating scales from different informants (i.e., parents, teachers, and self-report; Achenbach, McConaughy, & Howell, 1987; DiBartolo & Grills, 2006), thus a potential disadvantage to heavily relying on the BRIEF to examine ecological validity of tasks. Chevignard et al. (2012) argue two different types of measurements may assess different underlying constructs of executive function. Thus, tasks and questionnaires may provide more applicable information when taken together, rather than separately.

Measures of EF vary greatly; some measures use navigating and/or sorting (e.g., Key Search Task and Battersea Multitask Paradigm), whereas others examine function in an everyday task (e.g., BRIEF, The Children's Cooking Task). Findings from the current analysis suggest that many of the reviewed measures guided by verisimilitude occurred within a structured environment, therefore potentially reducing ecological validity. In addition, those measures guided by veridicality adapted aspects of traditional assessments to reflect more real-life scenarios (e.g., Virtual Continuous Performance Test, Gender Emotion Switch Task); however, we argue these adaptations still may not truly reflect performance in everyday life. This argument is analogous to Chevignard et al. (2012), who indicated many paper-pencil assessments of executive function were considered ecologically valid; however, performance under a restricted environment may not predict performance in everyday life. Therefore, executive function remains a difficult construct to measure and we should continue to develop innovative methods to better depict everyday life.

In addition, only one battery of tasks included in this review used parent report (i.e., BADS-C), whereas no other tasks accounted for parent insight. Parents provide the opportunity to elucidate a child's daily behavior within authentic contexts; the input of parent, caregiver, and teacher report during the assessment process is necessary to further understand EF in real-life contexts (Anderson, 2002). Overall, parent insight was underutilized when assessing child EF, and is paramount to measuring EF in an ecologically valid manner.

Finally, increased research is needed to understand the interplay between context and EF in children. The measures reviewed rarely aimed to measure EF and contextual features. Previous recommendations for increasing ecological validity in measurement emphasize the importance of environmental assessments (Olson, Jacobson, & Van Oot, 2013). By understanding the various EF demands within

myriad contexts, we may better pinpoint a child's EF abilities, and elucidate the relationship between measures and everyday life.

Limitations and Future Directions

In this scoping review, we aimed to understand the ecological validity of assessments for children. Although it was beyond the realm of this review, examining differences in reliability, validity, and standardization among measures may be useful to further understand assessments and ecological validity. Furthermore, including additional databases (e.g., Educational Resources Information Center [ERIC], Excerpta Medica database [EMBASE]) may have resulted in increased evidence reviewed in the current study. Moreover, expanding search terms (e.g., test, scale, instrument), and including other forms of literature (e.g., dissertations, theses, book chapters), may have expanded the search and resulted in additional literature. Finally, we only included studies regarding measurements with ecological validity components; however, analyzing all EF measures for ecological validity may have provided additional depth and understanding of EF in everyday life.

In addition, all reviewed assessments measured deficits in EF. A holistic assessment approach, however, would gather information of individual strengths and limitations during tasks that require EF. Children often use strategies and coping mechanisms to circumvent EF challenges; future research may contribute to an understanding of children's strengths and a basis for intervention.

Conclusion

Performance of daily tasks within natural contexts is central to OT practice, and the ideology of OT coincides with the ecological validity press in other fields. Dialogue reflecting issues of measurement is not new in the area of OT. However, discussion of ecologically valid measures remains less prominent. Coster and Khetani (2008) described issues and potential suggestions to increase the validity and usefulness of participation measures, and these issues align with those posing a risk to ecological validity. Specifically, Coster and Khetani (2008) depict the potential advantage of context-specific measures, as well as parent involvement when designing measures. Increased parental involvement, and creating measurements sensitive to different contexts, may be advantageous when creating more ecologically valid measures of EF. Thus, guiding OT measures by ecological validity principles may assist in generating future measures more sensitive to each child's executive function performance in daily life.

Declaration of Conflicting Interests

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References

- Achenbach, T. M., McConaughy, S. H., & Howell, C. T. (1987). Child/adolescent behavioral and emotional problems: Implications of cross-informant correlations for situational specificity. *Psychological Bulletin*, 101, 213-232. doi:10.1037/0033-2909.101.2.213
- Anderson, P. (2002). Assessment and development of executive function (EF) during childhood. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 8, 71-82. doi:10.1076/chin.8.2.71.8724
- Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8, 19-32. doi:10.1080/1364557032000119616
- Baron, I. S. (2007). Review of Behavioural Assessment of the Dysexecutive Syndrome for Children (BADS-C). *Child Neuropsychology*, 13, 539-542. doi:10.1080/09297040601112781
- Burgess, P. W., Alderman, N., Evans, J., Emslie, H., & Wilson, B. A. (1998). The ecological validity of tests of executive function. *Journal of the International Neuropsychological Society*, 4, 547-558.
- Burgess, P. W., Alderman, N., Forbes, C., Costello, A., Coates, L. M., Dawson, D. R., . . . Channon, S. (2006). The case for the development and use of "ecologically valid" measures of executive function in experimental and clinical neuropsychology. *Journal of the International Neuropsychological Society*, 12(2), 194-209. doi:10.1017/S1355617706060310
- Chaytor, N., & Schmitter-Edgecombe, M. (2003). The ecological validity of neuropsychological tests: A review of the literature on everyday cognitive skills. *Neuropsychology Review*, 13, 181-197. doi:10.1023/B:NERV.0000009483.91468.fb
- Chen, I.-C., Tsai, P.-L., Hsu, Y.-W., Ma, H.-I., & Lai, H.-A. (2013). Everyday memory in children with developmental coordination disorder. *Research in Developmental Disabilities*, 34(1), 687-694. Retrieved from <https://doi.org/10.1016/j.ridd.2012.09.012>
- Chevignard, M. P., Catroppa, C., Galvin, J., & Anderson, V. (2010). Development and evaluation of an ecological task to assess executive functioning post childhood TBI: The Children's Cooking Task. *Brain Impairment*, 11(2), 125-143. Retrieved from <https://doi.org/10.1375/brim.11.2.125>
- Chevignard, M. P., Soo, C., Galvin, J., Catroppa, C., & Eren, S. (2012). Ecological assessment of cognitive functions in children with acquired brain injury: A systematic review. *Brain Injury*, 26, 1033-1057. doi:10.3109/02699052.2012.666366
- Coster, W., & Khetani, M. A. (2008). Measuring participation of children with disabilities: Issues and challenges. *Disability and Rehabilitation*, 30, 639-648. doi:10.1080/09638280701400375
- Crone, E. A., & van der Molen, M. W. (2004). Developmental changes in real life decision making: performance on a gambling task previously shown to depend on the ventromedial prefrontal cortex. *Developmental Neuropsychology*, 25(3), 251-279. Retrieved from https://doi.org/10.1207/s15326942dn2503_2
- Davies, S. R., Field, A. R. J., Andersen, T., & Pestell, C. (2011). The ecological validity of the Rey-Osterrieth complex figure: Predicting everyday problems in children with neuropsychological disorders. *Journal of Clinical and Experimental Neuropsychology*, 33, 820-831. doi:10.1080/13803395.2011.574608
- de Almeida, R., Macedo, G., Lopes, E., & Monteiro, L. (2014). BADS-C instrument: An ecological perspective of the executive functions in children with attention deficit hyperactivity disorder. *Acta Neuropsychologica*, 12(3), 293-303. Retrieved from <http://search.ebscohost.com/login.aspx?direct=true&db=psyh&AN=2014-45117-002&site=ehost-live>
- de Vries, M., & Geurts, H. M. (2012). Cognitive flexibility in ASD; Task switching with emotional faces. *Journal of Autism and Developmental Disorders*, 42, 2558-2568. doi:10.1007/s10803-012-1512-1
- Díaz-Orueta, U., García-López, C., Crespo-Eguílaz, N., Sánchez-Carpintero, R., Climent, G., & Narbona, J. (2014). AULA virtual reality test as an attention measure: Convergent validity with Conners' Continuous Performance Test. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 20, 328-342. doi:10.1080/09297049.2013.792332
- DiBartolo, P. M., & Grills, A. E. (2006). Who is best at predicting children's anxiety in response to a social evaluative task? *Journal of Anxiety Disorders*, 20, 630-645. doi:10.1016/j.janxdis.2005.06.003
- Floyd, R. G., & Kirby, E. A. (2001). Psychometric properties of measures of behavioral inhibition with preschool-age children: Implications for assessment of children at risk for ADHD. *Journal of Attention Disorders*, 5(2), 79-91. doi:10.1177/108705470100500202
- Fournet, N., Roulin, J.-L., Monnier, C., Atzeni, T., Cosnefroy, O., Le Gall, D., . . . Roy, A. (2015). Multigroup confirmatory factor analysis and structural invariance with age of the Behavior Rating Inventory of Executive Function (BRIEF)-French version. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 21(3), 379-398. Retrieved from <https://doi.org/10.1080/09297049.2014.906569>
- Gerstle, M., Beebe, D. W., Drotar, D., Cassedy, A., & Marino, B. S. (2016). Executive Functioning and School Performance among Pediatric Survivors of Complex Congenital Heart Disease. *The Journal of Pediatrics*, 173, 154-159. Retrieved from <https://doi.org/10.1016/j.jpeds.2016.01.028>
- Gilboa, Y., Rosenblum, S., Fattal-Valevski, A., Toledano-Alhadeef, H., & Josman, N. (2014). Is there a relationship between executive functions and academic success in children with neurofibromatosis type 1? *Neuropsychological Rehabilitation*, 24(6), 918-935. Retrieved from <https://doi.org/10.1080/09602011.2014.920262>
- Goulden, L. G., & Silver, C. H. (2009). Concordance of the Children's Executive Functions Scale with established tests and parent rating scales. *Journal of Psychoeducational Assessment*, 27(6), 439-451. Retrieved from <https://doi.org/10.1177/0734282909335574>
- Isquith, P. K., Gioia, G. A., & Espy, K. A. (2004). Executive function in preschool children: examination through everyday behavior. *Developmental Neuropsychology*, 26(1), 403-422. Retrieved from https://doi.org/10.1207/s15326942dn2601_3
- Jaroslawska, A. J., Gathercole, S. E., Logie, M. R., & Holmes, J. (2016). Following instructions in a virtual school: Does working memory play a role? *Memory & Cognition*, 44, 580-589. doi:10.3758/s13421-015-0579-2

- Kenworthy, L., Yerys, B. E., Anthony, L. G., & Wallace, G. L. (2008). Understanding executive control in autism spectrum disorders in the lab and in the real world. *Neuropsychology Review*, 18, 320-338. doi:10.1007/s11065-008-9077-7
- Kronenberger, W. G., Castellanos, I., & Pisoni, D. B. (2016). Questionnaire-based assessment of executive functioning: Case studies. *Applied Neuropsychology: Child*, 1-11. Retrieved from <https://doi.org/10.1080/21622965.2016.1200976>
- Kvavilashvili, L., & Ellis, J. (2004). Ecological validity and the real-life / laboratory controversy in memory research: A critical and historical review. *History & Philosophy of Psychology*, 6, 59-80.
- Lawrence, V., Houghton, S., Douglas, G., Durkin, K., Whiting, K., & Tannock, R. (2004). Executive function and ADHD: A comparison of children's performance during neuropsychological testing and real-world activities. *Journal of Attention Disorders*, 7(3), 137-149. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/15260171>
- Lawrence, V., Houghton, S., Tannock, R., Douglas, G., Durkin, K., & Whiting, K. (2002). ADHD outside the laboratory: boys' executive function performance on tasks in videogame play and on a visit to the zoo. *Journal of Abnormal Child Psychology*, 30(5), 447-462. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/12403149>
- Levac, D., Colquhoun, H., & O'Brien, K. K. (2010). Scoping studies: Advancing the methodology. *Implementation Science*, 5 (69). doi:10.1186/1748-5908-5-69
- Li, Y., Cao, F., Shao, D., & Xue, J. (2014). Ecological assessment of executive functions in adolescents genetically at high risk for schizophrenia. *Comprehensive Psychiatry*, 55(6), 1350-1357. Retrieved from <https://doi.org/10.1016/j.compsych.2014.04.009>
- Lui, M., & Tannock, R. (2007). Working memory and inattentive behaviour in a community sample of children. *Behavioral and Brain Functions: BBF*, 3(12). Retrieved from <https://doi.org/10.1186/1744-9081-3-12>
- MacAllister, W. S., Vasserman, M., Coulehan, K., Hall, A. F., & Bender, H. A. (2016). Cognitive estimations as a measure of executive dysfunction in childhood epilepsy. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 22, 65-80. doi:10.1080/09297049.2014.967670
- Mackinlay, R., Charman, T., & Karmiloff-Smith, A. (2006). High functioning children with autism spectrum disorder: a novel test of multitasking. *Brain and Cognition*, 61(1), 14-24. Retrieved from <https://doi.org/10.1016/j.bandc.2005.12.006>
- McAuley, T., Chen, S., Goos, L., Schachar, R., & Crosbie, J. (2010). Is the behavior rating inventory of executive function more strongly associated with measures of impairment or executive function? *Journal of the International Neuropsychological Society*, 16, 495-505. doi:10.1017/S1355617710000093
- Nolin, P., Martin, C., & Bouchard, S. (2009). Assessment of inhibition deficits with the virtual classroom in children with traumatic brain injury: A pilot-study. *Studies in Health Technology and Informatics*, 144, 240-242. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/19592772>
- Olson, K., Jacobson, K. K., & Van Oot, P. (2013). Ecological validity of pediatric neuropsychological measures: Current state and future directions. *Applied Neuropsychology: Child*, 2, 17-23. doi:10.1080/21622965.2012.686330
- Payne, J. M., Hyman, S. L., Shores, E. A., & North, K. N. (2011). Assessment of executive function and attention in children with neurofibromatosis type 1: Relationships between cognitive measures and real-world behavior. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 17, 313-329. doi:10.1080/09297049.2010.542746
- Pnevmatikos, D., & Trikkaliotis, I. (2013). Intraindividual differences in executive functions during childhood: The role of emotions. *Journal of Experimental Child Psychology*, 115, 245-261. doi:10.1016/j.jecp.2013.01.010
- Portney, L. G., & Watkins, M. P. (2009). *Foundations of clinical research: Applications to practice* (3rd ed.). Philadelphia, PA: F. A. Davis.
- Pritchard, A. E., Kalback, S., McCurdy, M., & Capone, G. T. (2015). Executive functions among youth with Down Syndrome and co-existing neurobehavioural disorders. *Journal of Intellectual Disability Research: JIDR*, 59(12), 1130-1141. Retrieved from <https://doi.org/10.1111/jir.12217>
- Rosenblum, S., Frisch, C., Deutsch-Castel, T., & Josman, N. (2015). Daily functioning profile of children with attention deficit hyperactive disorder: A pilot study using an ecological assessment. *Neuropsychological Rehabilitation*, 25(3), 402-418. Retrieved from <https://doi.org/10.1080/09602011.2014.940980>
- Rosenthal, M., Wallace, G. L., Lawson, R., Wills, M. C., Dixon, E., Yerys, B. E., . . . Kenworthy, L. (2013). Impairments in real-world executive function increase from childhood to adolescence in autism spectrum disorders. *Neuropsychology*, 27(1), 13-18. Retrieved from <https://doi.org/10.1037/a0031299>
- Rosetti, M. F., Ulloa, R. E., Vargas-Vargas, I. L., Reyes-Zamorano, E., Palacios-Cruz, L., de la Peña, F., . . . Hudson, R. (2016). Evaluation of children with ADHD on the Ball-Search Field Task. *Scientific Reports*, 6, 19664. Retrieved from <https://doi.org/10.1038/srep19664>
- Roy, A., Allain, P., Roulin, J.-L., Fournet, N., & Le Gall, D. (2015). Ecological approach of executive functions using the Behavioural Assessment of the Dysexecutive Syndrome for Children (BADS-C): Developmental and validity study. *Journal of Clinical and Experimental Neuropsychology*, 37(9), 956-971. Retrieved from <https://doi.org/10.1080/13803395.2015.1072138>
- Siu, A. F. Y., & Zhou, Y. (2014). Behavioral Assessment of the Dysexecutive Syndrome for Children: An examination of clinical utility for children with attention-deficit hyperactivity disorder (ADHD). *Journal of Child Neurology*, 29(5), 608-616. Retrieved from <https://doi.org/10.1177/0883073813516191>
- Smithson, P. E., Kenworthy, L., Wills, M. C., Jarrett, M., Atmore, K., & Yerys, B. E. (2013). Real world executive control impairments in preschoolers with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, 43(8), 1967-1975. Retrieved from <https://doi.org/10.1007/s10803-012-1747-x>
- Solanto, M. V., Abikoff, H., Sonuga-Barke, E., Schachar, R., Logan, G. D., Wigal, T., . . . Turkel, E. (2001). The ecological validity of delay aversion and response inhibition as measures of impulsivity in AD/HD: A supplement to the NIMH multimodal treatment study of AD/HD. *Journal of Abnormal Child Psychology*, 29(3), 215-228. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/11411784>
- Tamm, L., Brenner, S. B., Bamberger, M. E., & Becker, S. P. (2016). Are sluggish cognitive tempo symptoms associated with executive

- functioning in preschoolers? *Child Neuropsychology*. Advance online publication. doi:10.1080/09297049.2016.1225707
- van den Bergh, S. F. W. M., Scheeren, A. M., Begeer, S., Koot, H. M., & Geurts, H. M. (2014). Age related differences of executive functioning problems in everyday life of children and adolescents in the autism spectrum. *Journal of Autism and Developmental Disorders*, 44(8), 1959-1971. Retrieved from <https://doi.org/10.1007/s10803-014-2071-4>
- Van Liefveringe, D., Sonuga-Barke, E., Van Broeck, N., Van Der Oord, S., Lemiere, J., & Danckaerts, M. (2016). A rating measure of ADHD-related neuropsychological impairment in children and adolescents: Data from the Cognition and Motivation in Everyday Life (CAMEL) Scale from population and clinical samples. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 1-19. Retrieved from <https://doi.org/10.1080/09297049.2016.1170798>
- Vriezen, E. R., & Pigott, S. E. (2002). The relationship between parental report on the BRIEF and performance-based measures of executive function in children with moderate to severe traumatic brain injury. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 8, 296-303. doi:10.1076/chin.8.4.296.13505
- Wehrle, F. M., Kaufmann, L., Benz, L. D., Huber, R., O'Gorman, R. L., Latal, B., . . . Hagmann, C. F. (2016). Very preterm adolescents show impaired performance with increasing demands in executive function tasks. *Early Human Development*, 92, 37-43. Retrieved from <https://doi.org/10.1016/j.earlhum-dev.2015.10.021>
- Weis, R., & Totten, S. J. (2004). Ecological validity of the Conners' Continuous Performance Test II in a school-based sample. *Journal of Psychoeducational Assessment*, 22, 47-61. doi:10.1177/073428290402200104
- White, S. J., Burgess, P. W., & Hill, E. L. (2009). Impairments on "open-ended" executive function tests in autism. *Autism Research: Official Journal of the International Society for Autism Research*, 2(3), 138-147. Retrieved from <https://doi.org/10.1002/aur.78>
- Whittingham, K., Bodimeade, H. L., Lloyd, O., & Boyd, R. N. (2014). Everyday psychological functioning in children with unilateral cerebral palsy: does executive functioning play a role? *Developmental Medicine and Child Neurology*, 56(6), 572-579. Retrieved from <https://doi.org/10.1111/dmcn.12374>
- Williams, D. L., Mazefsky, C. A., Walker, J. D., Minshew, N. J., & Goldstein, G. (2014). Associations between conceptual reasoning, problem solving, and adaptive ability in high-functioning autism. *Journal of Autism and Developmental Disorders*, 44(11), 2908-2920. Retrieved from <https://doi.org/10.1007/s10803-014-2190-y>
- Wolfe, K. R., Walsh, K. S., Reynolds, N. C., Mitchell, F., Reddy, A. T., Paltin, I., . . . Madan-Swain, A. (2013). Executive functions and social skills in survivors of pediatric brain tumor. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 19(4), 370-384. Retrieved from <https://doi.org/10.1080/09297049.2012.669470>